

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 1: The Earth: Geometry, Coordinates & Distances

Syllabus Source: CAE Oxford (pp. 1-42) | Pass Target: $\geq 90\%$ AviationExam

1. EARTH DIMENSIONS & COORDINATE SYSTEM

The Earth is an oblate spheroid (flattened at the poles, bulging at the equator) modeled by the WGS-84 ellipsoid:

- Dimensions & Flattening: Equatorial Diameter = 12,756 km (6,888 NM); Polar Diameter = 12,714 km (6,865 NM). Compression / Flattening = (Equatorial - Polar) / Equatorial = 1 / 298.25.
- Latitude (Parallels): Angular distance North or South of Equator (0° to 90°). Colatitude = $90^\circ - \text{Latitude}$. Parallels are SMALL CIRCLES (except the Equator, which is a Great Circle).
- Longitude (Meridians): Angular distance East or West of Prime Meridian (Greenwich 000°) (0° to 180°). All meridians are SEMI-GREAT CIRCLES.
- The Nautical Mile Definition: 1 Nautical Mile (NM) is legally defined as EXACTLY 1,852 METERS (6,076.1 ft). Physically, 1 NM = the length of 1 MINUTE OF ARC ALONG A MERIDIAN of latitude!

2. THE DEPARTURE FORMULA (DISTANCE ALONG A PARALLEL)

Because meridians converge toward the poles, the linear east-west distance between two meridians decreases as latitude increases:

- The Master Departure Formula: Departure (NM) = Change of Longitude (in minutes of arc) $\times \cos(\text{Latitude})$
- Change of Longitude (dlong): dlong (minutes) = Departure (NM) / $\cos(\text{Latitude})$ = Departure $\times \sec(\text{Latitude})$

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Departure Distance Along a Parallel

Problem: An aircraft flies due East along parallel $60^\circ 00' \text{N}$ from A (60°N , 010°W) to B (60°N , 020°E). What is the total flight distance in nautical miles? Step 1: Calculate total change of longitude (Chlong): - Crossing Greenwich meridian: Chlong = $10^\circ \text{W} + 20^\circ \text{E} = 30^\circ$ of longitude. Step 2: Convert degrees of longitude to minutes of arc: - Minutes = $30^\circ \times 60 \text{ min/degree} = 1,800$ minutes of longitude (or 1,800 NM at equator). Step 3: Apply latitude reduction cosine factor ($\cos 60^\circ = 0.50$): - Departure (NM) = Chlong (min) $\times \cos(\text{Latitude}) = 1,800 \times 0.50 = 900$ NM! Golden Rule: At 60° latitude, distance between meridians is exactly HALF of what it is at the equator!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating East-West Distance (Departure)

SCENARIO (Fundamental ATPL Navigation Drill): Two aircraft waypoints are located on the same parallel of latitude: - Waypoint A: 60°00'N 010°00'W - Waypoint B: 60°00'N 025°00'W QUESTION: What is the exact distance in Nautical Miles along the 60°N parallel between Waypoint A and Waypoint B? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Change in Longitude (dlong): - Longitude A = 010°W - Longitude B = 025°W - Both are in the same hemisphere (West), so subtract: dlong = 25° - 10° = 15°. Step 2: Convert degrees of longitude into minutes of arc: - Rule: 1 degree = 60 minutes of arc. - dlong (minutes) = 15° x 60 = 900 minutes of arc! Step 3: Recall the Master Departure Formula: - Formula: Departure (NM) = dlong (minutes) x cos(Latitude) Step 4: Plug in the numbers and calculate: - Latitude = 60°N. (Notice: cos(60°) = 0.50 exactly!). - Departure = 900 minutes x cos(60°) = 900 x 0.50 = 450 Nautical Miles! FINAL ANSWER: The distance along the 60°N parallel is exactly 450 Nautical Miles (half of the 900 NM it would be at the equator!).

[!] EXAM TRAP (AVIATIONEXAM) - The cos(60°) = 0.50 Shortcut Trap

- At 60° latitude (North or South), cos(60°) is EXACTLY 0.50! - This means at 60° latitude, 1° of longitude equals EXACTLY 30 NM (half of the 60 NM at the equator)! - Examiners LOVE 60° latitude questions because they can be solved instantly in your head without a calculator.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the distance in Nautical Miles along the parallel of 60°S between longitudes 020°E and 040°E?

- [A]: 1,200 NM
- [B]: 600 NM (20° x 60 min = 1,200 min x cos(60°) = 600 NM)
- [C]: 300 NM
- [D]: 1,852 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Change of longitude = 20° = 1,200 minutes. Departure = 1,200 x cos(60°) = 1,200 x 0.5 = 600 NM.

Q2: Which line on the Earth's surface is BOTH a parallel of latitude and a GREAT CIRCLE?

- [A]: Tropic of Cancer
- [B]: THE EQUATOR (0° Latitude)
- [C]: Greenwich Meridian
- [D]: Arctic Circle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Equator is the only parallel of latitude whose plane passes directly through the center of the Earth, making it a Great Circle.

Q3: What is the international legal definition of ONE NAUTICAL MILE?

- [A]: 6,000 feet
- [B]: Exactly 1,852 METERS (approx 6,076.1 ft)
- [C]: 1,609 meters

- [D]: 2,000 yards

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By international agreement in 1929 (adopted by ICAO), 1 NM is defined as exactly 1,852 meters.